Aarush Prasad Uli

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Education

University of Washington, Seattle — Expected June 2028
B.S. Computer Science, GPA: 3.78 / 4.0 — *Seattle, WA*

- Relevant Coursework: Data Structures, Algorithms, Systems Programming, Operating Systems, Databases, Hardware/Software Interface, Machine Learning, Deep Learning
- Selected for the Lavin Entrepreneurship Program (Autumn 2025 cohort), a competitive program within the Buerk Center for Entrepreneurship and Foster School of Business

Experience

DLESyM Undergraduate Research — Jan. 2026 – Present
Modeling & Simulation Research — UW Atmospheric Sciences & Geophysics — *Seattle, WA*

- Contributed to the Deep Learning Earth System Model (DLESyM), a GPU-accelerated Earth system simulation developed in collaboration with *NVIDIA*; advised by Prof. Dale Durran and PhD mentor Nathaniel Cresswell-Clay
- Engineered **Python** data pipelines processing **multi-TB** model output on HPC clusters, computing verification diagnostics (atmospheric blocking statistics, seasonal-cycle metrics, ENSO indices)
- Built reusable analysis and visualization tooling in **NumPy** and **PyTorch** to quantify model skill against reanalysis datasets; adopted by faculty and research staff for ongoing model validation

Four Elements Engineering — Sep. 2023 – Aug. 2025
Predictive Modeling Intern — *Redmond, WA*

- Built physics-based hydrological models in **Python** generating predictive stormwater runoff data for **3+** civil infrastructure projects across the Puget Sound region
- Developed internal tools automating field-measurement ingestion, data-quality validation, and recurring workflows, cutting manual processing time by ∼**80%**

Pack of Parts Robotics — Sep. 2023 – June 2025
Programming Sub-Team — Embedded Software & Controls — *Redmond, WA*

- Developed swerve-drive firmware in **Java** implementing closed-loop motor control with real-time encoder feedback under hard timing constraints; tuned PID loops for autonomous navigation
- Integrated an **OpenCV** vision pipeline for real-time target acquisition, fusing vision estimates with encoder state to drive an actuated arm, improving scoring efficiency by **55%**

Projects

OmniAge — Real-Time Triage Platform | *Spring Boot, Django, Kafka, Swift/SwiftUI, SQLite, Gemini API* — Sep. 2026 – Present

- Architected an event-driven backend for a native iOS triage platform: a **Django** REST service ingests and normalizes patient data from multiple sources and publishes record events to **Apache Kafka** topics, handling **100+** simulated records
- Built the triage engine as a **Spring Boot** microservice consuming Kafka events, maintaining a dynamic priority queue that re-ranks records by acuity and schedules follow-ups from **Gemini**-parsed free-text notes, reducing triage time by a projected ∼**40%**
- Designed RESTful APIs and the **SQLite** persistence layer; Kafka-driven queue updates keep a **SwiftUI** operator dashboard synchronized in real time, surfacing high-priority alerts as new data arrives

Tag! — Social Video Platform | *Flutter, AWS S3, AWS Bedrock, TwelveLabs* — Oct. 2025

- Built a Flutter social video app with user authentication, video submission, and content-verification workflows
- Designed an AWS-backed pipeline using **Amazon S3** for user and video storage, integrating **AWS Bedrock** and **TwelveLabs** for automated multimodal verification of submitted videos
- Presented the multimodal video-verification architecture to the *TwelveLabs* development team at its Multimodal Weekly Showcase

Technical Skills

Languages: Java, Python, C++, C, Swift, C#, Rust JavaScript/TypeScript, SQL
Backend & Data: Spring Boot, Django, Apache Kafka, REST APIs, SQLite, Linux
Cloud & Tools: AWS (S3, Bedrock), Git, Linux, React, Flutter, SwiftUI
ML & Data Science: PyTorch, NumPy, OpenCV, Gemini API
Spoken Languages: English (native), Japanese (fluent), Spanish (fluent), Hindi